EXPERIMENT 8

```
% Define parameters

clc;

 clear;

 close all;

N = 50; % Number of samples f0 = 0.1;

 % Frequency component 1 f1 = 0.3;

% Frequency component 2 n = 0:N-1;

 % Sample index

% Define the signal x = cos(2pif0n) + sin(2pif1n);

% Compute DTFT omega = linspace(-pi, pi, 1000);

 % Frequency range X = x * exp(-1j * (n.' * omega)); % DTFT calculation

% Plot the magnitude of DTFT figure;

plot(omega/pi, abs(X));

xlabel('Normalized Frequency (\omega/\pi)');

 ylabel('|X(e^{j\omega})|'); title('Magnitude of DTFT');

grid on;

% Plot the phase of DTFT figure;

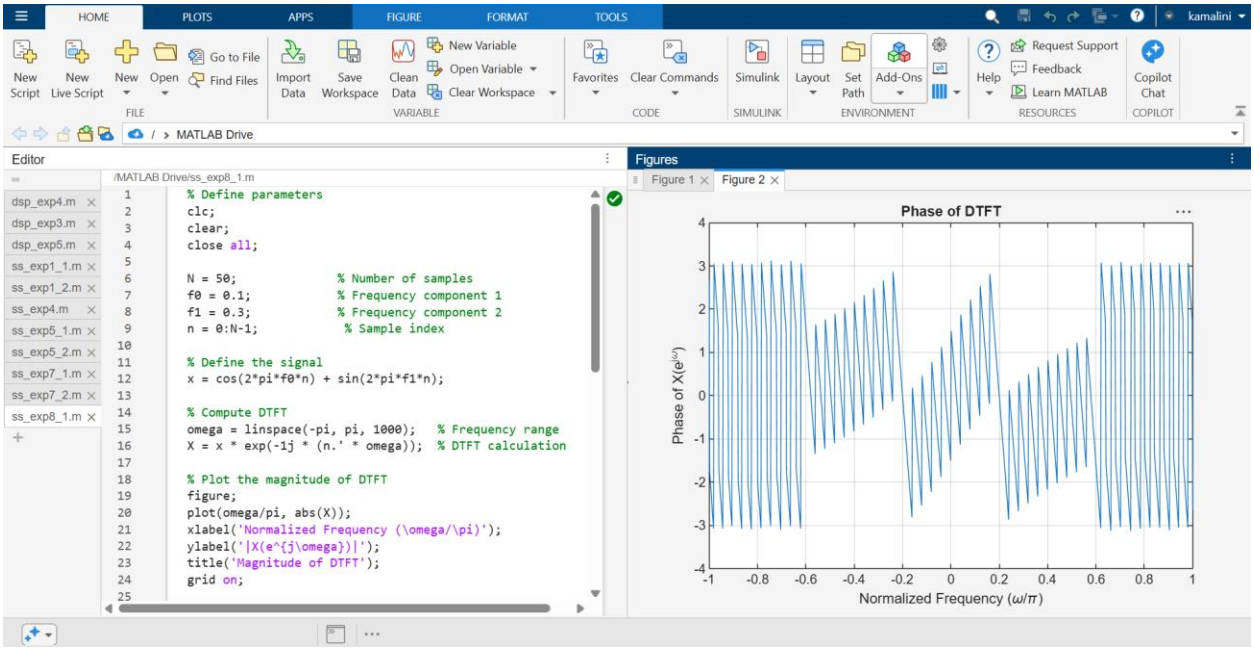
 plot(omega/pi, angle(X));

 xlabel('Normalized Frequency (\omega/\pi)');

 ylabel('Phase of X(e^{j\omega})');

 title('Phase of DTFT');
```

grid on;



2:

% Define parameters clc; clear; close all;

N = 31;

% Total number of samples n = 0:N-1;

% Sample index n1 = 10;

% Start of rectangular pulse n2 = 20;

% End of rectangular pulse

% Define the rectangular pulse x[n] x = zeros(1, N);

% Initialize the signal with zeros x(n1+1:n2+1) = 1;

% Set values between n1 and n2 to 1

% Compute DTFT omega = linspace(-pi, pi, 1000); % Frequency range X = x * exp(-1j * (n.' * omega)); % DTFT calculation

% Plot the magnitude of DTFT figure;

```
plot(omega/pi, abs(X));
xlabel('Normalized Frequency (\omega/\pi)');
ylabel('|X(e^{j\omega})|');
title('Magnitude of DTFT of Rectangular Pulse');
% Plot the phase of DTFT figure;
plot(omega/pi, angle(X));
xlabel('Normalized Frequency (\omega/\pi)');
ylabel('Phase of X(e^{j\omega})');
title('Phase of DTFT of Rectangular Pulse');
```

